Data Science Project Scoping

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Team Information

Team Name: Data Insight Drivers

Name of team members:	Kampamba Chanda	Lehlogonolo Mbiza
Student number of team members:	14366798	19064722
Signatures		
Date	13/03/2026	13/03/2026

Project Information

Project Name: Integrating High-Resolution Imagery and Machine Learning for Small and Informal-Area Population Estimation in Melusi (Atteridgeville, Pretoria)

Project Lead: Samy Katumba

Signatures	
Dates	13/03/2026

Project Information

1. **Project Name**
   Integrating High-Resolution Imagery and Machine Learning for Small and Informal-Area Population Estimation in Malusi (Atteridgeville, Pretoria)

2. **Organisation Name:**
   University of Pretoria

3. **Project Description:**
   Develop a model to estimating the population in an area (specifically Melusi Atteridgeville, Pretoria) using drone aerial photographs and an ensemble U-Net to Random Forest Regression architecture, to complement traditional census methods.

4. **Who are the agencies/departments that will need to be involved?**
   Department of Geography, Geoinformatics and Meteorology (GGM) & Gauteng Research Triangle (GRT)

5. **Who are the individuals in these organizations that are stakeholders? What are their role?**
   Samy Katumba – University of Pretoria – Project Lead

Project Goals

6. **Goals (in order of priority)**

   What are you maximizing or minimizing?
   Are there any constraints (budget, resources, etc.)?

Goal 1: Identify a suitable deep-learning technique to detect and extract building footprints/profiles from high-resolution imagery.	**Goal 2:** Construct a suitable machine learning model for estimating population size and/or density using extracted building footprints/profiles and building density metrics, and a recent household survey conducted in Melusi.
Constraints: Core data sources split through multimodal (RGB, LiDAR, GCP Shape, and GCP reference) 3cm resolution may require significant compute	**Constraints:** Dependent on quality and completeness of the household survey data.

Goal 3: Evaluate the validity and performance of both deep-learning and machine models constructed in this research study.	**Goal 4:** Conclude by proposing a methodological framework that can be used as a blueprint for replicating the procedures adopted to other informal settlements in South Africa.
Constraints: Ground truth data availability for validation; no independent population count to validate against beyond the household survey.	**Constraints:** Generalisation of framework to extend to other settlements and layouts. Scaling might be dependent on population assumptions (e.g., do informal settlements have different demography from each other?)

Project Actions

7. **Actions**

Action 1: Detect and map building footprints in Melusi from drone imagery	**Action 2:** **Understand data and data connectors**	**Action 3:** Estimate segment population from building density metrics
Who is taking the action? Data Insight Drivers	**Who is taking the action?** **Data Insight Drivers**	**Who is taking the action?** Data Insight Drivers
What/Who is it being taken on? The drone imagery of Melusi	**What/Who is it being taken on?** Provided data sets	**What/Who is it being taken on?** The drone imagery of Melusi
How often is the decision to take this action made? Once of for the project and every other time when the done on a new settlement	**How often is the decision to take this action made?** Once of for the project and every other time when the done on a new settlement	**How often is the decision to take this action made?** Once of for the project and every other time when the done on a new settlement
What channels are used to take this action Digital	**What channels are used to take this action** Digital	**What channels are used to take this action** Digital
Other questions about the action	**Other questions about the action**	**Other questions about the action**

Data Information

8. **Data**
   A. What Data is available internally at the partner?

Data Source 1 Drone Imagery (ECW)	**Data Source 2** Drone Imagery (TIFF)	**Data Source 3** GPS Coordinates
What does it contain? RGB aerial image		**What does it contain?** 4 Reference GCP's and GNSS quality metrics
What level of granularity? 3cm special resolution		**What level of granularity?** Point level
How frequently is it collected/updated after it's captured? Data is collected 3 times a year		**How frequently is it collected/updated after it's captured?** Once off
Does it have unique identifiers that can be linked to other data sources? Yes, georeferenced with 4 GCP's		**Does it have unique identifiers that can be linked to other data sources?** Yes, georeferenced with 4 GCP's
Who's the internal owner of the data? GCM Department, UP		**Who's the internal owner of the data?** GCM Department, UP
How is it stored? Cloud filing system		**How is it stored?** Cloud filing system
Other DJI Zenmuse L1 sensor Large file for training		**Other** Text files committed to GitHub repo

Data Source 4 GPS Shape Files	**Data Source 5** LiDAR Point Cloud
What does it contain? 4 Geospatial Vector GCP's	**What does it contain?** 3D Point cloud representing terrain heights
What level of granularity? Point level	**What level of granularity?** Point level
How frequently is it collected/updated after it's captured? Every census period	**How frequently is it collected/updated after it's captured?** Every census period
Does it have unique identifiers that can be linked to other data sources? Yes, linked to drone imagery via spatial coordinates	**Does it have unique identifiers that can be linked to other data sources?** Yes, spatially referenced to align with drone imagery
Who's the internal owner of the data? GCM Department, UP	**Who's the internal owner of the data?** GCM Department, UP
How is it stored? Cloud filing system	**How is it stored?** Cloud filing system
Other Lo29 South African coordinate system stored in .prj file	**Other** Can be used to distinguish single vs multi-storey dwellings for population estimation.

	Can also be used to determine statistical probability of dwelling being located at those coordinates

B. What data can you get externally and /or from public sources?

Data Source 1: Google Open Building footprints	**Data Source 2:** OpenStreetMap Building footprints
What does it contain? Polygons of building footprints from satellite imagery	**What does it contain?** Community managed building outlines
What level of granularity? Individual building level	**What level of granularity?** Individual building level
How frequently is it collected/updated after it's captured? Periodically updated	**How frequently is it collected/updated after it's captured?** Continuously managed by volunteers
Does it have unique identifiers that can be linked to other data sources? Yes, spatially referenced	**Does it have unique identifiers that can be linked to other data sources?** Yes, spatially referenced
Who's the internal owner of the data? Google (Open License)	**Who's the internal owner of the data?** OpenStreetMap (Open License)
How is it stored? Online, publicly accessible via api call)	**How is it stored?** Online, publicly accessible via api call)
Other: Lower resolution imagery used, can be used to validate U-Net outputs at courser scale.	**Other** Flat 2D building shapes that covers Atteridgeville. Should be mapped to Melusi to assist U-Net training

Analysis Goals and Needs

9. Analysis

Analysis 1: Exploratory Data Analysis of all datasets	**Analysis 2:** Evaluation of the U-Net model	**Analysis 3:** Evaluation of the Random Forest Regression model
Analysis type: Descriptive/ Exploratory	**Analysis type:** Predictive	**Analysis type:** Predictive
Which action will this analysis inform? Action 1. Understanding data quality and structure will inform how we link the data to each other.	**Which action will this analysis inform?** Action 1 and action 2 in that an accurate footprint extraction will be the basis for model	**Which action will this analysis inform?** Action 2 and action 3 in that the model can estimate a population close to the survey actual population.
How will you validate this analysis? Visual validation using compressed ecw and QGIS to confirm alignment	**How will you validate this analysis?** Model quantitative metrics: IoU, precision, recall, F1-score	**How will you validate this analysis?** Cross referencing the estimated population with the population from the survey by checking the MSE between true and estimated